

ITE 19

Study Guide

ICT Concepts

Topic learning Outcome

- a. Review knowledge in ICT Concepts
- b. Comprehend ICT Concepts to apply in real-life settings

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1. Database Concepts

This section is designed to provide a foundational understanding of databases and their role in information management. By the end of this module, you will understand what a database is, how it differs from simple file storage, the core components of a relational database, and the basics of how to interact with one using SQL.

Learning Objective:

Upon completion of this topic, you will be able to:

- **Define** a database and **explain** its advantages over a simple file

- system. • **Differentiate** between data, information, and knowledge.
- **Identify** and **describe** the key components of a relational database: tables, records, fields, and keys.
- **Explain** the purpose of a Database Management System (DBMS).
- **Write** basic SQL commands to query and manipulate data.

1.1. Introduction to Databases

What is a Database (DB)?

An organized collection of logically related data, designed to be efficiently stored, retrieved, and managed.

Database vs. Simple File System

Aspect	Simple File System	Database
Data Redundancy	High (same data stored in multiple files)	Low (data is integrated)
Data Consistency	Difficult to maintain	Enforced through rules
Data Sharing	Difficult	Easy and controlled
Security	Limited to OS file permissions	Granular user access control
Complex Queries	Very difficult, requires custom programs	Easy, using SQL

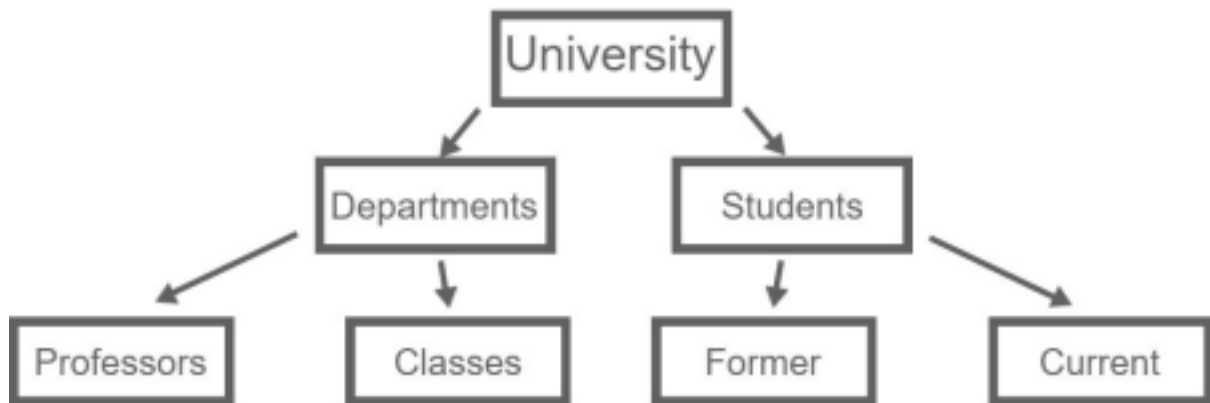
Data vs. Information vs. Knowledge

Data	Information	Knowledge
Raw, unorganized facts and figures (e.g., 42, Smith, 100).	Data that has been processed and given context to be meaningful (e.g., Employee ID 42, John Smith, has a salary of \$100,000).	The application of information to make decisions or solve problems (e.g., John Smith is the highest-paid employee in his department).

1.2. Database Models

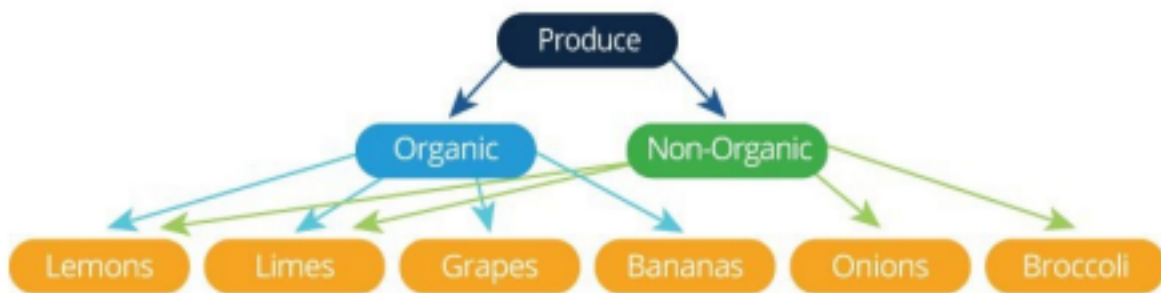
A database model defines the logical structure of a database.

- **Hierarchical Model:** Data is organized in a tree-like structure (parent-child relationships). Rarely used today.



- **Network Model:** An extension of the hierarchical model allowing a child to have multiple parents. Complex and rarely used.

Network Database Model



The network model has parent-child relationships, but allows many-to-many relationships.

- **Relational Model:** The most common model. Data is organized in **tables (relations)** with rows and columns. This manual focuses on this model.

1.3. Relational Database Concepts

Table also called Relation

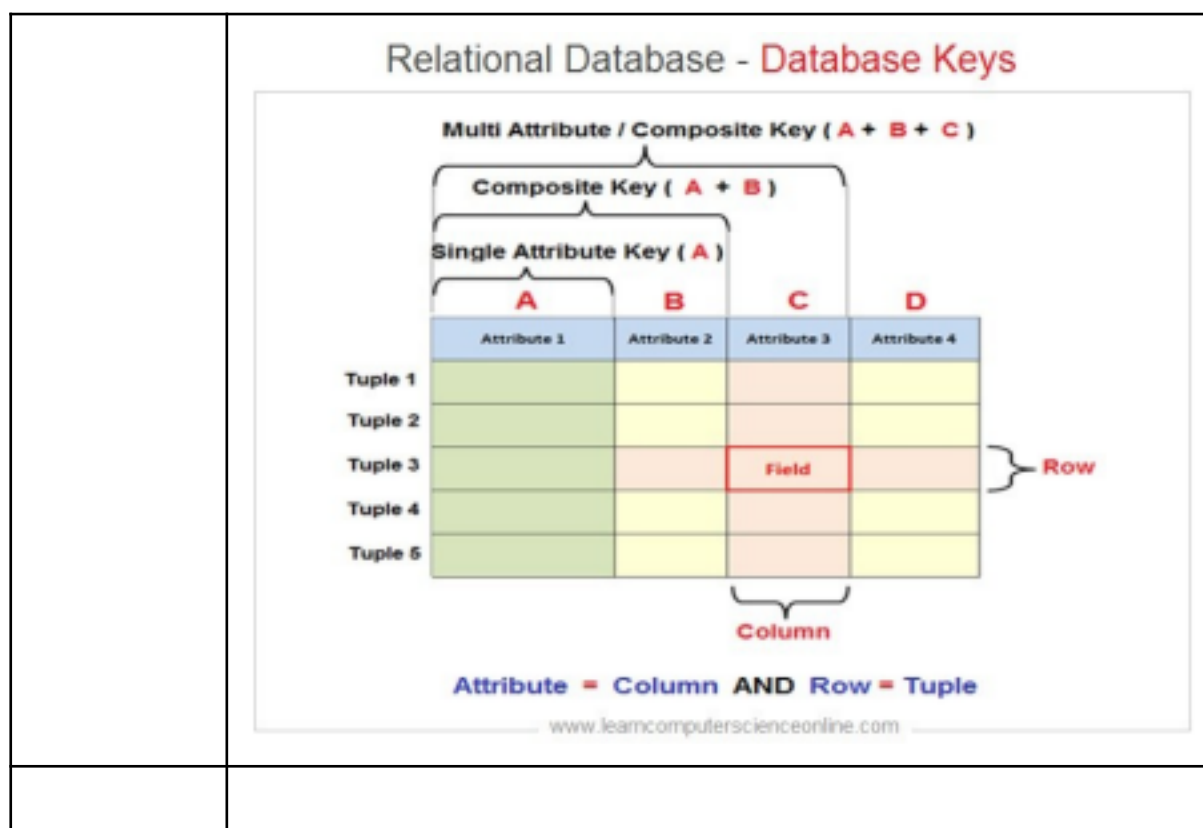
Primary Key Domain
Eg: NOT NULL © guru99.com

CustomerID	CustomerName	Status
1	Google	Active
2	Amazon	Active
3	Apple	Inactive

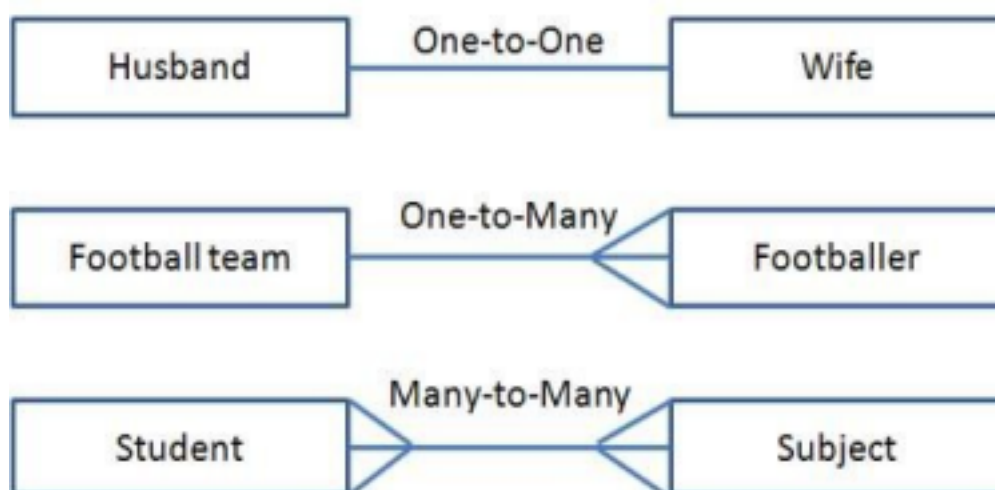
Tuple OR Row
Total # of rows is **Cardinality**

Column OR Attributes
Total # of column is **Degree**

Table (Relation)	A collection of related data held in a structured format. Think of it as a spreadsheet. Example: A Customers table, a Products table, an Orders table.
Record (Tuple)	A single, horizontal row in a table that represents a complete instance of an entity. Example: One row in the Customers table containing all data for a single customer.
Field (Attribute):	A single, vertical column in a table that represents a specific property of an entity. Example: The CustomerID, FirstName, LastName columns in the Customers table.
Keys	Special fields used to establish relationships and ensure uniqueness 🚩 Primary Key: A unique identifier for each record in a table (e.g., CustomerID in the Customers table). It cannot be null. 🚩 Foreign Key: A field in one table that uniquely identifies a record in another table. It creates a link between two tables (e.g., CustomerID in the Orders table is a foreign key linking back to the Customers table). 🚩 Candidate Key: Any field or combination of fields that could be chosen as the primary key.



1.3.1. Relationships: One-to-one, One-to-many, Many-to-many



Relationships define how data in one table is linked to data in another table.

- **One-to-One (1:1):** One record in Table A relates to only one record in Table B, and vice versa. (e.g., A Employees table and a ParkingSpots table).
- **One-to-Many (1:M):** One record in Table A relates to many records in Table B. This is the most common relationship.
 - **Example:** One customer (in Customers) can place many orders (in Orders).
- **Many-to-Many (M:N):** Many records in Table A relate to many records in Table B. This requires a **junction table** to break it into two one-to-many relationships.

- **Example:** An Orders table and a Products table. One order can contain many products, and one product can be in many orders. The junction table OrderDetails would contain fields like OrderID, ProductID, and Quantity.

1.4. Database Management Systems (DBMS)

What is a DBMS?

Software that interacts with end-users, applications, and the database itself to capture and analyze data. It serves as an interface between the database and its users.

• Functions of a DBMS:

- Data storage, retrieval, and update.
- Managing database security and authorization.
- Ensuring data integrity (rules to ensure data is accurate and consistent).
- Managing concurrency control (allowing multiple users to access data simultaneously without conflict).
- Providing backup and recovery.

- **Examples of DBMS Software:** MySQL, PostgreSQL, Microsoft SQL Server, Oracle Database, SQLite.

1.5. Structured Query Language (SQL)

SQL is the standard language used to communicate with a relational database.

- **Data Definition Language (DDL):** Defines and modifies database structures (e.g., CREATE, ALTER, DROP tables).
- **Data Manipulation Language (DML):** Manages data within tables (e.g., SELECT, INSERT, UPDATE, DELETE).

Basic SQL Commands:

SELECT Retrieves data from one or more tables.	SELECT FirstName, LastName FROM Customers WHERE Country = 'USA';
INSERT Adds new records to a table	INSERT INTO Customers (FirstName, LastName) VALUES ('John', 'Doe');

UPDATE Modifies existing records in a table	UPDATE Customers SET Country = 'UK' WHERE CustomerID = 42;
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DELETE Removes records from a table	DELETE FROM Customers WHERE CustomerID = 100;
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1.6. Summary

Databases provide a structured, efficient, and secure way to manage large amounts of data, overcoming the limitations of simple file systems. The relational model, organized in tables with relationships defined by keys, is the most prevalent. A DBMS is the software that powers the database, and SQL is the universal language used to interact with it.

1.7. Practice Exercises

